

Plant vascular system

The transport system of a plant is made up of two sets of vessels—xylem and phloem. Xylem carries water around the plant. Its stiff tubes run from the roots, up the stem, to the leaves. Phloem carries the sugar made in the leaves to the rest of the plant in the form of dissolved sucrose. Both types of vessel are made from columns of cells with openings at either end that form continuous pipes along which liquids can flow.

More than **100 million tons** of sugar are extracted from the sap stored in the phloem tubes of sugar cane **every year**.

REAL WORLD

Giant redwood

The largest trees in the world, such as these giant redwoods of California, USA, grow to around 361 ft (110 m) tall. Scientists estimate that this is about the maximum height for a tree, since the pressure needed to pump a continuous column of water any higher would cause the water to pull itself apart inside the tree, and never reach the top.

